

Phase 3 Video Course

Course Title: HelixCode Phase 3 - Advanced AI Development Features

Target Audience: Developers using HelixCode for AI-assisted development

Total Duration: ~120 minutes (12 videos, 8-12 minutes each)

Difficulty: Intermediate to Advanced

12 Videos

120 Minutes

6 Modules

Intermediate

Course Structure

Module 1: Introduction and Overview (20 minutes)

Video 1.1: Phase 3 Overview (8 min)

- What's new in Phase 3
- The 5 core features overview
- How they work together
- Use cases and benefits
- Demo preview

Video 1.2: Getting Started (12 min)

- Installation and setup
- Basic configuration
- First session creation
- Quick win examples
- Common patterns

Module 2: Session Management (25 minutes)

Video 2.1: Session Management Fundamentals (10 min)

- What are sessions?
- Session modes (planning, building, testing, refactoring, debugging, deployment)
- Session lifecycle (create, start, pause, resume, complete)
- Status tracking
- Demo: Creating and managing sessions

Video 2.2: Advanced Session Management (15 min)

- Project association
- Tagging and organization
- Session queries and filtering

- Session history and statistics
- Export/import sessions
- Demo: Multi-project workflow

Module 3: Memory and Context (25 minutes)

Video 3.1: Memory System Basics (10 min)

- Messages and conversations
- Role-based messaging (user, assistant, system)
- Conversation management
- Token tracking
- Demo: Building conversation history

Video 3.2: Advanced Memory Features (15 min)

- Search and filtering
- Message limits and trimming
- Conversation statistics
- Export/import conversations
- Best practices for memory management
- Demo: Managing long conversations

Module 4: State Persistence (20 minutes)

Video 4.1: State Persistence Fundamentals (10 min)

- Why persistence matters
- Save and load operations
- Storage formats (JSON, compact, GZIP)
- Atomic writes and data integrity
- Demo: Saving and restoring state

Video 4.2: Advanced Persistence (10 min)

- Auto-save configuration
- Backup and restore
- Migration strategies
- Performance optimization
- Demo: Recovery workflows

Module 5: Template System (25 minutes)

Video 5.1: Template Basics (10 min)

- Template types (code, prompt, workflow, documentation, email, custom)
- Variable substitution syntax
- Creating templates
- Built-in templates
- Demo: Simple template creation and rendering

Video 5.2: Advanced Templates (15 min)

- Template manager features
- Search and filtering
- File operations
- Export/import templates
- Template libraries
- Demo: Building a template library

Module 6: Integration and Real-World Usage (25 minutes)

Video 6.1: Integration Patterns (12 min)

- Session + Memory integration
- Templates + Conversations
- Full workflow examples
- Best practices
- Demo: AI-assisted feature development

Video 6.2: Advanced Workflows (13 min)

- Multi-session development
- Code review workflow
- Debugging sessions
- Template-based generation
- Performance tips
- Demo: Complete development workflow

Video Scripts

Detailed scripts are provided in the repository:

- [01_phase3_overview.md](#)
- [02_getting_started.md](#)
- [03_session_fundamentals.md](#)
- [VIDEOS_04_12_COMPLETE_SCRIPTS.md](#) (Videos 4-12)

Learning Objectives

By the end of this course, students will be able to:

- ✓ Create and manage development sessions effectively
- ✓ Build and maintain conversation history with the memory system
- ✓ Persist and recover application state reliably
- ✓ Create and use templates for code generation
- ✓ Integrate all Phase 3 features in real-world workflows
- ✓ Optimize performance and manage resources
- ✓ Troubleshoot common issues
- ✓ Implement advanced integration patterns

Prerequisites

- Basic knowledge of HelixCode
- Familiarity with Go programming
- Understanding of AI-assisted development concepts
- Completion of Phase 1 & 2 courses (recommended)

Assessment

Quizzes

- End of each module quiz (5-10 questions)
- Final comprehensive quiz (20 questions)

Projects

- Build a complete development workflow
- Create a custom template library
- Implement session recovery system
- Multi-project integration

Certification

Upon completion:

- HelixCode Phase 3 Certified Developer
- Digital badge
- Certificate of completion

Phase 3 Features

API Reference

Example Projects